

Religious Organization Calendar App — Design Document

Sources

- Conversation analysis — schema design and UI wireframe from prior messages
- [CodeInterpreter] — SQL schema generation

Core Thesis

A comprehensive calendar application for religious organizations must serve both administrative efficiency and spiritual community needs, combining standard event management with liturgical awareness, volunteer coordination, and multi-campus support.

[Draft] 1. Executive Summary

This document presents the formal technical design for a calendar application tailored to religious organizations. The application supports multi-denomination, multi-campus environments and addresses unique requirements including liturgical season tracking, scripture/lectionary integration, volunteer scheduling, and role-based visibility for congregation members and leadership.

[Draft] 2. Problem Statement

Religious organizations manage complex schedules spanning worship services, sacramental events, ministry groups, volunteer coordination, outreach programs, and administrative meetings. Existing general-purpose calendar tools lack support for liturgical context, congregation-specific roles, volunteer gap identification, and spiritual content integration (e.g., scripture readings, prayer focuses). This application bridges that gap.

[Draft] 3. Functional Requirements

3.1 Core Event Management

- Create, read, update, delete events with full recurrence support (RFC 5545 RRULE)
- Multi-campus and room/facility booking
- Event categories with color coding
- RSVP/registration with capacity limits and waitlisting
- Cancellation and weather alert notifications

3.2 Liturgical & Spiritual Features

- Liturgical season tracking with color theming
- Daily/weekly scripture reading assignments (lectionary support)
- Prayer focus scheduling

3.3 People & Groups

- Member directory with role-based access (pastor, admin, leader, volunteer, member)
- Ministry groups and small group management
- Group membership tracking

3.4 Volunteer Management

- Role definitions with min/max volunteer counts
- Assignment scheduling per event
- Status tracking (assigned, confirmed, declined, swapped)
- Gap identification with visual alerts

3.5 Notifications & Alerts

- Event reminders, cancellations, and updates
- Weather alerts
- Volunteer gap notifications
- Scheduled send support

3.6 Visibility & Access Control

- Public events (visible to all, including website visitors)
- Members-only events
- Leaders-only events (board meetings, staff planning)

[Draft] 4. Database Schema

4.1 Entity Relationship Overview

Entity	Key Relationships	Purpose
organizations	Parent of all entities	Multi-org/denomination support
campuses	→ organizations	Multi-site locations
rooms	→ campuses	Facility booking
members	→ organizations	People directory & auth
events	→ organizations, categories, campuses, rooms	Core calendar entries
event_categories	→ organizations	Color-coded groupings
event_registrations	→ events, members	RSVP tracking
liturgical_seasons	→ organizations	Seasonal context & theming
ministry_groups	→ organizations, campuses	Small groups & ministries
group_members	→ ministry_groups, members	Group roster
scripture_readings	→ organizations, events	Lectionary / daily readings
prayer_focuses	→ organizations	Weekly prayer themes
volunteer_roles	→ organizations	Role definitions

Entity	Key Relationships	Purpose
volunteer_assignments	→ events, volunteer_roles, members	Scheduling
notifications	→ organizations, events	Alerts & reminders

4.2 Full SQL Schema

(Include complete SQL DDL with all CREATE TABLE statements, constraints, and indexes as specified in Section 4.2 of the approved schema artifact from conversation history)

[Draft] 5. UI Wireframes

5.1 Screen Inventory

Screen	Purpose	Key Components
Dashboard / Home	At-a-glance overview	Liturgical banner, today's scripture, upcoming events, prayer focus, alerts
Calendar (Month/Week/Day/List)	Browse and filter events	Color-coded grid, campus and category filters, legend
Event Detail (Flyout)	View full event information	Date/time, location, amenities, scripture, volunteers, actions
Volunteer Schedule	Manage service assignments	Role × event matrix, status indicators, gap alerts
Create/Edit Event (Modal)	Add or modify events	Form with all event fields, recurrence, visibility, RSVP options

5.2 Dashboard / Home Screen

The dashboard provides immediate situational awareness:

- **Liturgical Season Banner** — Current season name, color, and week number
- **Today's Scripture** — Assigned reading for the day
- **Upcoming Events** — Next 7 days with category colors, time, location, and amenity icons
- **Prayer Focus** — Current weekly prayer theme
- **Alerts** — Volunteer gaps, schedule changes, and announcements

5.3 Calendar View

A full monthly grid with:

- Color-coded event dots per category
- Toggle between Month, Week, Day, and List views
- Filters for campus and category
- Quick-add event button
- Today indicator

5.4 Event Detail Flyout

A slide-in panel showing:

- Category badge and event title
- Date, time, recurrence pattern
- Location (campus + room)
- Description
- Amenity icons (livestream, accessible, childcare, parking)
- Scripture readings for the service
- Volunteer roster with gap warnings
- Contact information
- Action buttons: Watch Live, Add to Calendar, Edit

5.5 Volunteer Schedule View

A matrix layout:

- Rows = volunteer roles (Greeter, Usher, Sound, Nursery, Music)
- Columns = upcoming services/events
- Cells = assigned volunteer name + status icon
- Visual warning for unfilled slots
- Actions: Send Reminders, Auto-Schedule, Export

5.6 Create/Edit Event Modal

A comprehensive form with:

- Title, category, event type
- Start/end date-time with all-day and recurrence options
- Campus and room selection
- Description rich text
- Options: RSVP, capacity, childcare, livestream, visibility
- Contact person assignment
- Save as Draft / Publish workflow

[Draft] 6. Design Principles

Principle	Implementation
Liturgical Awareness	Season banner and color always visible; theming adapts to liturgical calendar
At-a-Glance Scanning	Color-coded categories, iconography for amenities and status
Role-Based Views	Admins see edit/assign controls; members see RSVP/join actions
Multi-Campus Support	Global campus filter; events scoped to specific locations
Accessibility-First	Icons for wheelchair access, childcare, livestream; WCAG 2.1 AA compliance
Mobile-Responsive	Sidebar collapses; stacked layout for small screens
Proactive Gap Surfacing	⚠ indicators for unfilled volunteer roles and schedule conflicts

[Draft] 7. Technical Considerations

7.1 Recurrence Handling

- Uses RFC 5545 RRULE format for maximum flexibility
- Supports: daily, weekly, bi-weekly, monthly, annual, custom patterns
- Recurrence exceptions stored separately to handle cancellations of individual occurrences

7.2 Timezone Support

- Organization-level timezone setting
- All timestamps stored in UTC; displayed in local timezone
- Critical for multi-campus orgs spanning time zones

7.3 Performance

- Indexed on org_id + start_datetime for fast calendar queries
- Partial index on recurring events to optimize expansion
- Pagination for list views; lazy-load for calendar grids

7.4 Security & Privacy

- Role-based access control at the application layer
- Visibility field on events enforces who can see what
- Member contact info only visible to authorized roles

[Draft] 8. Future Enhancements

- Mobile native app (iOS/Android) with push notifications
- Calendar sync (iCal, Google Calendar, Outlook export)
- Giving/donation integration tied to events
- Attendance tracking with QR code check-in
- Sermon/media library linked to worship events
- Multi-language support for diverse congregations
- Analytics dashboard (attendance trends, volunteer participation rates)